

Day 20: Complete Chi-Square Workflow: Python Code Automation Explained

1. Approach 1: The Easy Manual Math Method (Step-by-Step Logic)

English: This approach breaks down the complete Chi-Square formula step-by-step using basic variables. It avoids loops and advanced collections to ensure that the core mathematical steps are completely visible and clear in the code structure.

Hindi: Yeh approach pure Chi-Square formula ko basic variables ka use karke ek-ek step mein todti hai. Isme loops ya complex collections ka use nahi kiya gaya hai taaki code ke structure mein math ka core logic ekdum saaf aur crystal clear dikhai de.

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# =====
# METHOD 1: EKDUM EASY MANUAL CODE (BINA LOOP KE PURE MATH)
# =====

print("--- STEP 1: Observed Data (Jo Table Mein Diya Hai) ---")
t_rings = 50
t_chains = 150
l_rings = 150
l_chains = 50

print("Tourist Rings:", t_rings, "| Tourist Chains:", t_chains)
print("Local Rings:", l_rings, "| Local Chains:", l_chains)

print("\n--- STEP 2: Totals Nikalna (Plus Karke) ---")
tourist_total = t_rings + t_chains      # 50 + 150 = 200
local_total = l_rings + l_chains        # 150 + 50 = 200
rings_total = t_rings + l_rings         # 50 + 150 = 200
chains_total = t_chains + l_chains      # 150 + 50 = 200
grand_total = tourist_total + local_total # 200 + 200 = 400

print("\n--- STEP 3: Degree of Freedom (df) ---")
df = (2 - 1) * (2 - 1)                 # 1 * 1 = 1
print("Degree of Freedom (df) =", df)

print("\n--- STEP 4: Expected Values Nikalna ---")
exp_t_rings = (tourist_total * rings_total) / grand_total # 100
exp_t_chains = (tourist_total * chains_total) / grand_total # 100
exp_l_rings = (local_total * rings_total) / grand_total # 100
exp_l_chains = (local_total * chains_total) / grand_total # 100

print("\n--- STEP 5: Chi-Square Score ( $\chi^2$ ) ---")
part1 = ((t_rings - exp_t_rings) ** 2) / exp_t_rings      # 25
part2 = ((t_chains - exp_t_chains) ** 2) / exp_t_chains # 25
part3 = ((l_rings - exp_l_rings) ** 2) / exp_l_rings      # 25
part4 = ((l_chains - exp_l_chains) ** 2) / exp_l_chains  # 25

chi_square_score = part1 + part2 + part3 + part4          # 25 + 25 + 25 + 25 = 100
print("Final Calculated Chi-Square Score =", chi_square_score)

print("\n--- STEP 6: Percentage Nikalna (Library Se) ---")
from scipy import stats
p_value_table = stats.chi2.sf(chi_square_score, df)
percentage_table = p_value_table * 100
print("Table Data Percentage:", percentage_table, "%")

```

2. Approach 2: The Industrial Library Method (1-Line Magical Solution)

English: This approach leverages advanced library automation to perform all horizontal, vertical, expected cell evaluations, and degree of freedom scaling instantly using an integrated statistical engine.

Hindi: Yeh approach advanced library automation ka use karti hai jisse horizontal totals, vertical totals, expected cells, aur degree of freedom scaling bina kisi manual math ke ek single statistical engine se turant solve ho jata hai.

```
# =====  
# METHOD 2: PROFESSIONAL INDUSTRIAL LIBRARY METHOD (TURANT SOLUTION)  
# =====  
  
from scipy import stats  
  
# STEP 1: Sirf Table Ka Raw Data Matrix Input Karo  
observed_data = [[50, 150], [150, 50]]  
  
# STEP 2: Single Line Library Function Jo Sab Solve Kar Dega  
chi2, p_val, df, expected = stats.chi2_contingency(observed_data)  
  
# STEP 3: Results Print Karo Percentage Me Badal Kar  
print("Calculated Chi-Square Score:", chi2)  
print("Degree of Freedom (df):", df)  
print("Exact P-Value Percentage:", p_val * 100, "%")
```

3. Structural Differences & Learning Takeaway

English: Use the manual approach when you need to embed the exact mathematical formula components directly into your working memory. Transition to the library method when building clean, production-grade automated pipelines.

Hindi Summary: Jab dimaag mein pure mathematical steps aur formulae ko bariki se bithaana ho, toh bina loop ka manual tareeqa sabsay behtar hai. Aur jab kaam ko fast track par daal kar real-time automated data pipelines banana ho, toh library function ka use kiya jata hai.